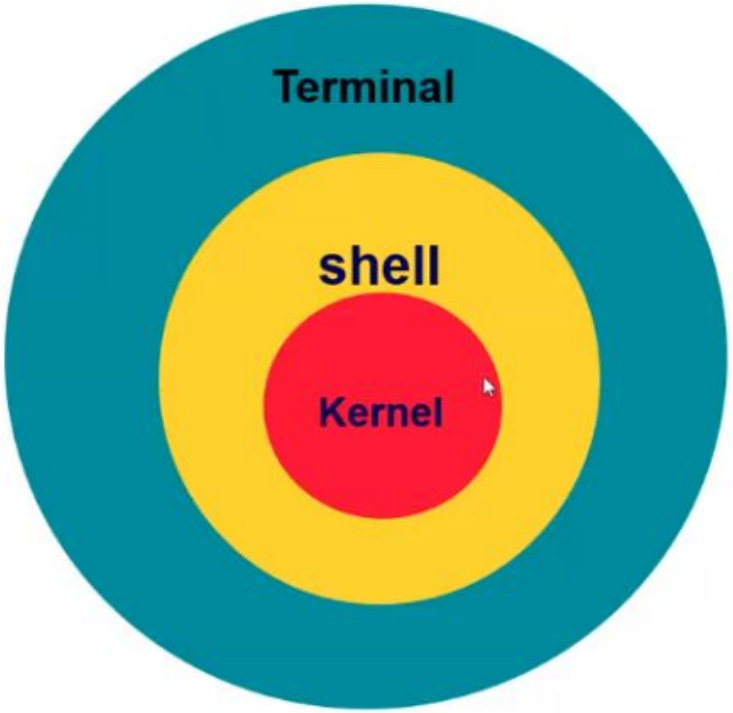
Introduction to Linux



Introduction to Linux

- Difference between shell and kernel

Parameters	Shell	Kernel
Basics	A shell is basically an interface present between the kernel and the user.	A kernel is the very core of a typical OS.
Uses and Purpose	A shell allows all of its users to establish communication with the kernel.	A kernel functions to control all the tasks that come with a system.
Layer of OS	The shell forms the outer layer of the operating system.	The kernel forms the inner layer of the operating system

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Text Editors (vi / vim)

- Default editor in all UNIX operating systems.
- vi in Linux is usually vim (vi improved)
- Supports syntax highlighting.
- vi three basic modes
  1. Command mode
     - Perform commands to delete, copy, …
  2. Edit mode
     - Enter text into the file
  3. Last line mode
     - To access it, enter a colon (:) while in the command mode

Text Files Utilities

- **Viewing file contents**
  - **cat file-name**
  - **more file-name or less file-name**
    - Scrolling keys for the more command
    - Space bar: moves forward on screen
    - b: move back one screen
    - /string: search forward for pattern
    - :n find the next occurrence
    - :q quit and return to the shell prompt
- **head -n file-name**
- **tail file-name**

Wildcard Characters

- **Asterisk(*): represents 0 or more character**
  - ls f*
- **Question mark(?) character represents any single character**
  - ls file?
- **Square bracket([ ]): represent a range of characters for a single character position**
  - ls [a-f]*

On



Browsing and exploring filesystem

- **cd change directory**
  - cd dir-name go to dir
  - cd – go previous
  - cd . go to current
  - cd .. go up
- **pwd print current working directory**
- **ls list files in current directory**

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Copying and moving files

- cp copy files and directories
  - **cp source destination**
- mv move & rename files & directories
  - **mv source destination**
    - If in same dir, it's considered rename.
    - If to another partition or directory, it's considered move.

Lab1

- mkdir parent1
- cd parent1
- pwd
- touch file1
- touch file2
- rm file1
- mv file1 file3
- mkdir child1
- mkdir child2
- mv file1 child1
- rm -r child1
- cd child2
- cd ..

Text Editors

1. **GUI-based editors**
   - **gedit**
2. **Terminal-based editors**
   - **nano**
     - Simplest text-based editor.
   - **vi / vim**
     - More powerful but complex

TextEditing (vim notes)

- /mohammed -> this search for mohammed
- :3 -> this goes to line 3
- i -> lets you insert into the file
- cat -> print a file content.

wildcards

- (*) -> get all the items with 0 character or more
- ? -> search for certain number of characters depending on the # of ? exists.
- [a-f] -> search on characters in that range.

general commnads

- head -2 filename
    - gets the first 2 lines in the filename

permissions

file or dir owner group others

```
-         ---   ---   ---
```

> to give premission we wirte (chmod)

> to add we use chmod+

> to remove we use chmod-

> u -> for user o -> owner g -> grp

piping

m3naha eny 3auz a5od el output bta3 command abaseh le command tany fa bst5mdo cat file1 | sort keda hwa hytb3 el content bta3 file1 w b3d keda haya5od el output da w ye3mlo sort.

Redirection

Redirection

bta5od el output bta3 command mo3ayn w t7oto fe file.

> command > file -> redirect the normal outputs

> command 2> file -> redirect the output of errors

> command &> file -> redirect both.

searching

grep -> we search for certain words in a files or directories.

greb -v key -> keda hygeb kol el files el mfhash el key da.

archiving

> **tar** this command is used to archive the files, like zipping them in the

Shell Scripting

- we mainly uses the scripting for automation.
- also we use it for data extraction by which we mean that we get certain informations from our files
- also used to monitor the system
- finally used for running some testcases.
- we can write all commands inside the script.
- we have many type of shell one of them is Cshell and another one is Bash

MakeFiles

1. bn3ml file 3ady w nektb gowah esm el target "hello"
2. bnsave w ne2fl
3. bnktb make w nektb esm el target
    1. make hello

1. make hello

- lazm el esm bta3o ykon esmo makefile aw MAKEFILE
- lw 3ndk kaza target fa mmkn t5ly wahed ynfz wahed abl ma ytnfz hwa fa sa3tha da esmo dependencies
    - all : one two
        - keda all dependenet on one and two
        - fa keda lama tegy tnfz all haynfz one el awl w b3den ynfz two.
- lw btktb command lazm tnzl satr w te3ml tap
- lw btktb dependency fa byb2o gamb b3d

makefile parameter

- bast5dmo variable gowa el makefile
- x:=1 keda 3rft x w kemtha 1
- b3d keda btst5dm el esm w 34an taccess el value bnst5dm el $x

why make files?

- lw 3ndk files ktera w 7auz trtb el compilation bta3hom.
- b7es en kol file by-depend 3la wahed tany, fa ana gowa el make file baa bartbhom b-mzagy
- mn esmo aslun hwa by3ml make lel hagat, akno bybny structure kda.

General notes

- mv f1 f2
    - this move all the content of file1 into file2
    - this means that it overwrites the content not appending
    - and then removes f1.
- mv f1 dir1
    - here we move file f1 into directory dir1
- cd (no inputs)
    - it goes to the home.

- mv f1 dir1
    - here we move file f1 into directory dir1
- cd (no inputs)
    - it goes to the home.
- cp f1 f2
    - this copy the content of f1 into f2.
- there are two admins who can change the permissions either the root or the owner of the file.
- **group** is a file on linux defined inside it a group of people who share the same access.
- sudo is for the root and some other users which have the sudo permission.
-

> linux is open source

> open source means that anybody can take a look at and modify it.

> introduction to linux notes:
>
> > Kernel: el mo5 bta3 el operating system el ms2ol eno ye2sm el hardware bta3 el system kolo w ye2dr yedy awamr lel hardware wl memory
>
> > Shell: is an interface between the user and the kernel, the user write the commands and the kernel applyies them.
>
> > Terminal: the GUI on which the user write the commands and the shell output its results on.

Command Line

- cd
  - change directory
- ls
  - list

> we have 3 ways to get help about commands

- man
- whatis
- --help option

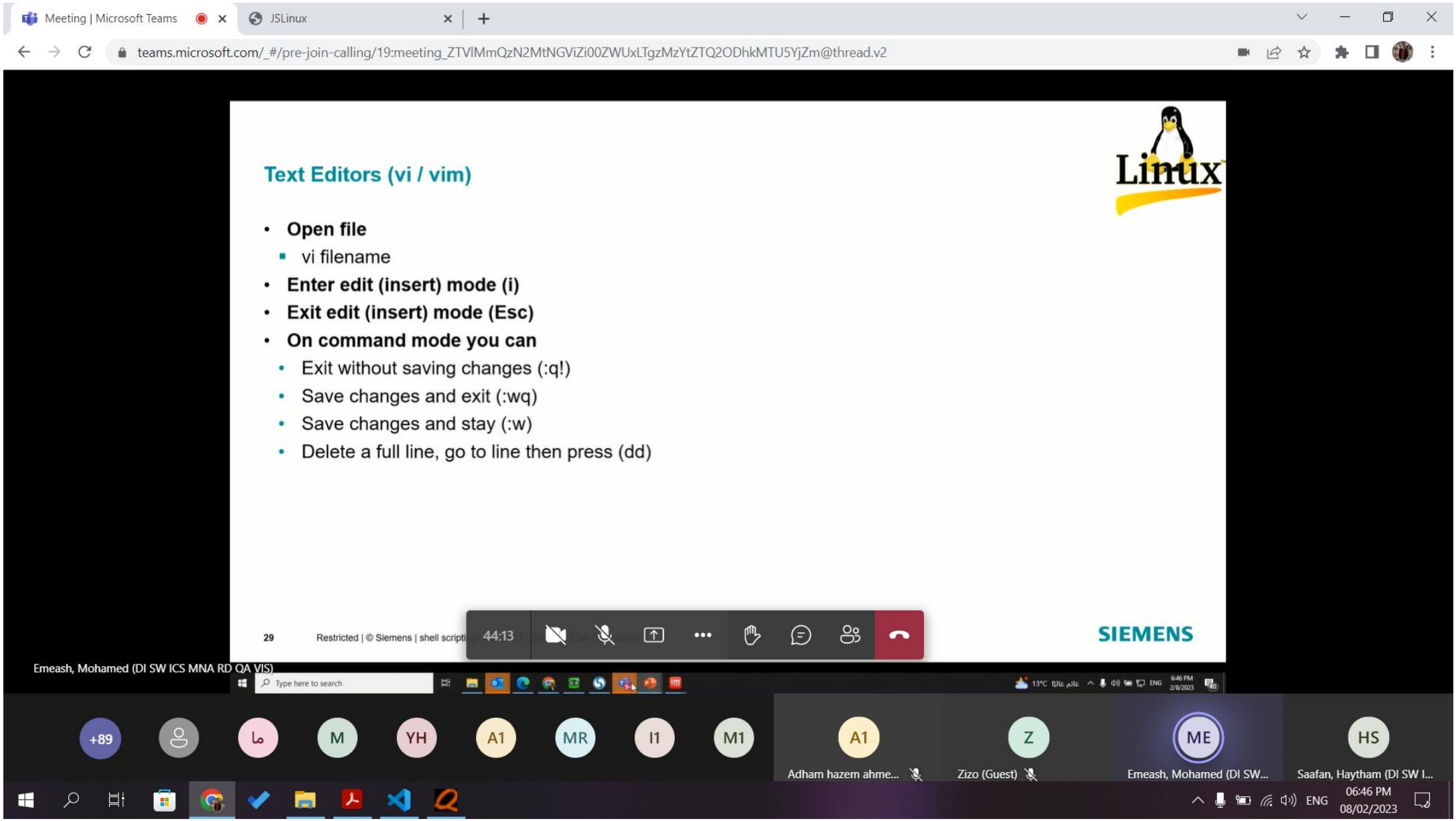
> man contains all the commands inside it you can find any command you wanna use.

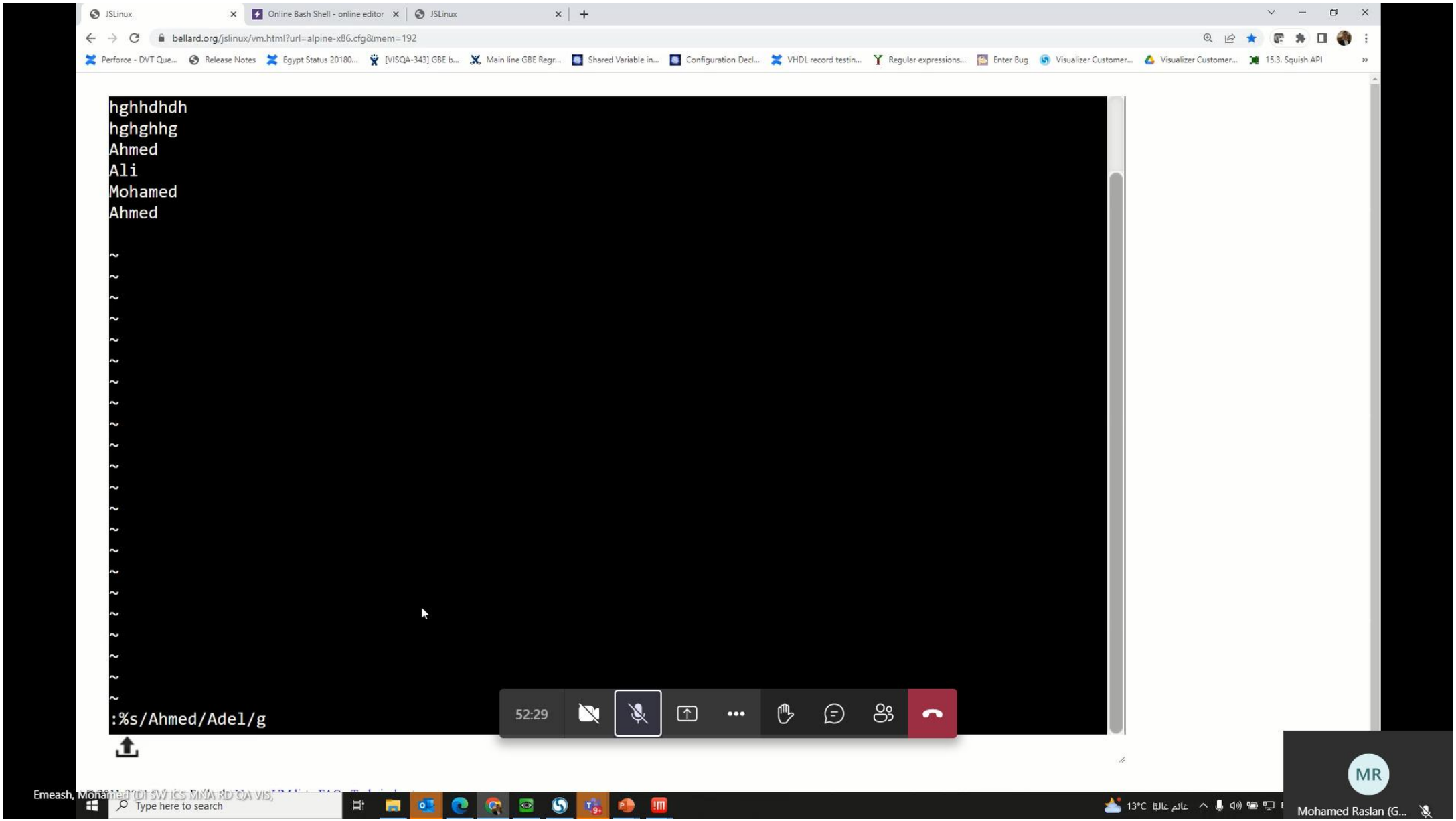
creating files

> touch -> create a file mkdir -> create a directory

browsing and exploring file systems

> cd cd- cd.


Meeting | Microsoft Teams
JSLinux
teams.microsoft.com/_#/pre-join-calling/19:meeting_ZTVlMmQzN2MtNGViZi00ZWUxLTgzMzYtZTQ2ODhkMTU5YjZm@thread.v2
Text Editors (vi / vim)
Linux
Open file
vi filename
Enter edit (insert) mode (i)
Exit edit (insert) mode (Esc)
On command mode you can
Exit without saving changes (:q!)
Save changes and exit (:wq)
Save changes and stay (:w)
Delete a full line, go to line then press (dd)
29
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44:13
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M
YH
A1
MR
I1
M1
A1
Adham hazem ahme...
Z
Zizo (Guest)
ME
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HS
Saafan, Haytham (DI SW I...
ENG
06:46 PM
08/02/2023


JSLinux
Online Bash Shell - online editor
JSLinux
bellard.org/jslinux/vm.html?url=alpine-x86.cfg&mem=192
hghhdhdh
hghghhg
Ahmed
Ali
Mohamed
Ahmed
:%s/Ahmed/Adel/g
52:29
MR
Mohamed Raslan (G...
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Text Files Utilities



- **Viewing file contents**
  - **cat file-name**
  - **more file-name or less file-name**
    - Scrolling keys for the more command
    - Space bar: moves forward on screen
    - b: move back one screen
    - /string: search forward for pattern
    - :n find the next occurrence
    - :q quit and return to the shell prompt
- **head -n file-name**
- **tail file-name**

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MA
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Wildcard Characters



- **Asterisk(*): represents 0 or more character**
  - ls f*
- **Question mark(?) character represents any single character**
  - ls file?
- **Square bracket([ ]): represent a range of characters for a single character position**
  - ls [a-f]*

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Managing and securing files and permissions



- Each file has an owner and assigned to a group.
- Linux allows users to set permissions on files and directories to protect them.
- Permissions are assigned to
  - – File owner
  - – Members of the group the file assigned to
  - – All other users
- Permissions can only be changed by the owner and root.

Unmute
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Saafan, Haytham (DI ...

Managing and securing files and permissions



Permissions on files and directories		
permission	File Access	Directory Access
Read	You can display file contents and copy the file	You can list the directory contents
Write	You can modify the file contents	you can add and delete files in the directory
Execute	You can execute the file if it is an executable	You can use the cd command to access the directory

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Managing and securing files and permissions



- **To change permissions on file**
  - chmod
- **Symbolic mode**
  - chmod u+x file
  - chmod g+w file
  - chmod g-r file
- **Numeric (octal) mode**
  - Read permission has a value of 4
  - Write permission has a value of 2
  - Execute permission has a value of 1
  - chmod 555 file
  - chmod 755 file

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Piping



- **Piping makes use of interprocess communication.**
- **Piping is to take the output of first command as an input to the second command.**
- **Piping two commands format:**
  - First command | second command
- **Examples**
  - cat file-name | sort
  - cat file-name | tail -n 2

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Redirection



- It’s to redirect the output channel to a text file to use as log or any other usage.
- There’s two main output channels
- Standard output channel
  - command > file
- Standard error channel
  - command 2> file
- To redirect both channels
  - command &> file

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Mazen (Guest)

Searching inside text files



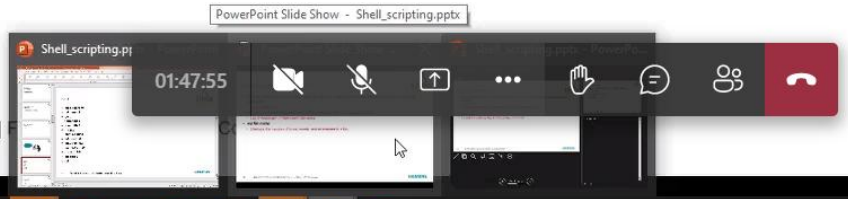
- **To make a horizontal search in a text file**
  - grep (selection)
- **It prints all the lines matching a pattern**
  - grep search-pattern file-name
- **To print all lines without the matching pattern**
  - grep -v pattern file-name
- **It can be used with piping**
  - cat file-name | grep pattern

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Zizo (Guest)

Searching inside text files (cont'd)



- **To find a definite column from a text file which is organized in fields delimited with a predefined character.**
  - cut (projection)
- **Remove sections from each line of file and print only the required field**
  - cut -d "delimiter" -f "field-num" file-name
- **wc file-name**
  - Displays the number of lines, words, and characters in a file.



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Archiving & Compression



tar command archives files to and extracts files from a single file called a tar file.

- **Options:**
  - c: create a new tar file
  - t: list table of content
  - x: extracts files from the tar command
  - f: specify the archive file
  - v: verbose mode

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Dina Atef joined the group via invite link
Zizo (Guest)
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Archiving Files



Examples

- **Creating an archive**
  - tar -cvf file.tar file1 file2
    - file1
    - file2
- **Viewing an archive**
  - tar -tf file.tar
    - file1
    - file2
- **Extracting files from archive**
  - tar -xvf file.tar
    - file1
    - file2

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Shell Scripting

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Zizo (Guest)

Shell Scripting



- **Why do we need shell scripts ?**
  - There are many reasons to write shell scripts
    1. **Automation**:
       - To avoid repetitive work and task automation.
       - Compile src files
       - Run testcases
    2. **Data extraction**
       - Extract errors and define pass criteria
    3. **System monitoring.**
    4. **System admins use the shell scripting for routine backups.**

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Shell Scripting

- **Why do we need shell scripts ?**
  - It's very useful skill for every engineer in the DIC industry:

Digital Design Engineer



Verification Engineer



QA/DEV Engineer



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Basic Script Architecture



- **The first line is**
  - #!(shell-path)
- **It may contain one or more of these**
  - variables definition
  - read variables from user
  - loops
  - conditional statements
  - print output to terminal
  - pipe & redirection to other files

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Creating and Running Scripts



- **Scripts are normal text files with x permission**
  - touch scriptname
  - chmod 777 scriptname
  - chmod u+x scriptname
- **To run a script:**
  - ./scriptname
  - bash scriptname

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Variables



- **Define a variable**
  - variablename=VALUE
- **Print a variable to screen**
  - echo $variablename
- **read take a value from the user to a variable**
  - read variablename
  - read -p “string” variablename

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Bash Comparison Operators



- Equal (=)
- Not equal (!=)
- Less than (-lt)
- Greater than (-gt)

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Conditional statements



- **if statement**

```
if [ condition ]
then
    ###things to do
elif [ another-cond. ]
then
    ###things to do
else
    ###things to do
fi
```


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Conditional statements (cont'd)



- case statement

```
case $variablename in
 value_1)
   ###things to do;;
 value_2)
    ###things to do;;
 *)
    ###default action;;
 esac
```


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Loops



- for loop

```
for VARIABLE in ARRAY
do
    Command1
    Command2
    Command3
done
```


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Lab2



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Lab2

- **User Login Example:**

```
#!/bin/bash
user_name="mohamed"
password="17"
read -p "User Name: " login_name
read -p "Password: " login_pass
echo ""
if [ $login_name = $user_name ];then
    if [ $login_pass = $password ];then
        echo "Now you are logged in....."
    else
        echo "Error, wrong password, try again"
    fi
else
    echo "Error, user name doesn't exist"
fi
```


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02:02:16

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Quiz

- Write a bash script for calculator program which has 3 inputs (first number, second number and operation) , the operation should be one of 4 operations (+,-,x,/) after that print the output based on the operation.
- Tip: use the case statement for operation selection.

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Solution



- **Calculator Example:**

```
#!/bin/bash
read –p "Enter First Number: " f_num
read –p "Enter Second Number: " s_num
read –p "Enter Operation(+,-,x,/): " op
case $op in
+)
   echo "$f_num + $s_num = $[$f_num+$s_num]";;
-)
   echo "$f_num $s_num = $[$f_num-$s_num]";;
x)
  echo "$f_num x $s_num = $[$f_num*$s_num]";;
/)
  echo "$f_num / $s_num = $[$f_num/$s_num]";;
esac
```


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Zizo (Guest)

Content



1. Makefile structure
2. Makefile parameters
3. Makefile dependences
4. Usage of Makefile

Youssef Hossam
02:29:20
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Makefile Target



- **Makefile structure**

```
target1: prerequisites
<TAB>   command
        command

target2: prerequisites
<TAB>   command
        command

target3: prerequisites
<TAB>   command
        command
```



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Makefile Example:



- **Hello world example:**
  - touch makefile
  - vi makefile

```
hello:
        echo "Hello, World“
```

  - make hello

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Makefile Example:




```
all: one two three
one:
        touch one
two:
        touch two
three:
        touch three
clean:
        rm -f one two three
```


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Makefile Example



- **Exmaple1:**
  - make one
  - make all
- **Example2:**
  - make one
  - make clean
  - make all

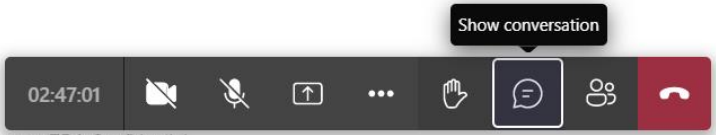
```
all: one two three
one:
        touch one
two:
        touch two
three:
        touch three
clean:
        rm -f one two three
```


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Ibrahim nasr (Guest)

Makefile Parameters:



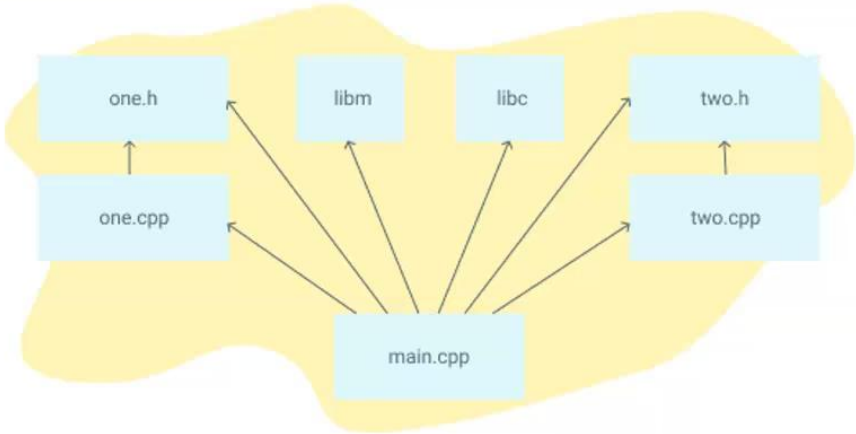

```
x:=1
hello:
        echo $x
```



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Why Makefile ?

- Lots of source files
- How to mange them all.
- Compiling is complicated.
- Solution: make



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